📄 AI ASSISTANT AGENT SYSTEM

A Smart Multi-Source AI Agent using RAG + Web + LLM
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🧠 INTRODUCTION

This project is an intelligent AI agent designed to answer user queries by dynamically selecting the most appropriate source of information.

The system integrates three core capabilities:

- Retrieval-Augmented Generation (RAG)
- Web Search Integration
- Direct Language Model (LLM) Responses

The goal is to provide accurate, context-aware, and real-time responses while minimizing hallucination.

⚙️ SYSTEM ARCHITECTURE

The AI agent follows a layered architecture:

1. **User Input Layer**
   Accepts user queries via text or voice input.
2. **Decision Layer (Router)**
   Determines how to answer the query by selecting:
   - RAG (document-based answers)
   - Web Search (real-time information)
   - LLM (general conversation)
3. **Knowledge Layer (RAG System)**
   - Documents are converted into embeddings
   - Stored in a FAISS vector database
   - Relevant chunks are retrieved using similarity search
4. **Response Generation Layer**
   - Uses LLaMA 3 via Groq API
   - Combines retrieved context with reasoning
5. **Output Layer**
   - Displays response in UI
   - Optional voice output

🔍 RETRIEVAL-AUGMENTED GENERATION (RAG)

RAG improves response accuracy by grounding answers in stored documents.

Process:

1. Documents are loaded and split into smaller chunks
2. Each chunk is converted into vector embeddings
3. Stored in FAISS vector database
4. On query:
    - Top relevant chunks are retrieved
    - Passed to LLM as context

Benefits:

- Reduces hallucination
- Provides document-based answers
- Enables semantic search

🌐 WEB SEARCH INTEGRATION

The system uses DuckDuckGo for real-time information retrieval.

Triggered when:

- Queries involve current events
- Keywords like “latest”, “news”, “today” are present

Workflow:

- Search results are fetched
- Passed to LLM for summarization
- Clean answer returned to user

🤖 LANGUAGE MODEL (LLM)

The system uses **LLaMA 3 (via Groq API)**.

Capabilities:

- Natural conversation
- Explanation and reasoning
- Summarization
- General knowledge responses

Used when:

- Query does not require document context
- Query is conversational

🎤 VOICE INTERACTION

The system supports voice-based interaction.

Features:

- Speech-to-text using `speech_recognition`
- Text-to-speech using `gTTS`

This allows users to interact with the system hands-free.

📂 DOCUMENT HANDLING

Users can upload PDF documents dynamically.

Behavior:

- Uploaded PDFs are processed into embeddings
- Merged into the existing vector database
- Expands the system's knowledge in real time

🧠 SMART ROUTING LOGIC

The agent automatically selects the best response mode:

- **RAG** → when query relates to documents
- **Web Search** → for real-time or external queries
- **LLM** → for general conversation

This makes the system adaptive and intelligent.

🚀 FEATURES SUMMARY

- Multi-source intelligent responses
- Real-time web search capability
- Document-based reasoning (RAG)
- Voice input and output
- Dynamic knowledge expansion
- Clean and interactive UI (Streamlit)

🔮 FUTURE ENHANCEMENTS

- AI-based routing (instead of keyword-based)
- Source citation in responses
- Multi-document ranking

- Persistent memory across sessions
- Deployment as API/service

🏁 CONCLUSION

This AI agent demonstrates how combining RAG, LLM, and web search creates a powerful intelligent system.

It mimics real-world AI assistants by dynamically selecting the best source of information for each query, ensuring both accuracy and flexibility.